KAIZEN MATHS

Lesson starter

Matrices: Operations, Determinants and Inverses

Determinants, Inverses and Multiplication

Name _____

S1

Find $\begin{bmatrix} -2 & 0 \\ 1 & -3 \end{bmatrix}\begin{bmatrix} -3 & 4 \\ 5 & -3 \end{bmatrix}$.

S2

Find the determinant of $\begin{bmatrix} -2 & 6 \\ 7 & 5 \end{bmatrix}$.

S3

Find the inverse of $\begin{bmatrix} -4 & 6 \\ 0 & 2 \end{bmatrix}$.

S4

Find the determinant of $\begin{bmatrix} -1 & 2 & 0 \\ 5 & 5 & -3 \\ 4 & -1 & -4 \end{bmatrix}$.

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KAIZEN MATHS

Exit ticket

Matrices: Operations, Determinants and Inverses

Determinants, Inverses and Multiplication

Name _____

E1

Find $\begin{bmatrix} -1 & -3 \\ 5 & -1 \end{bmatrix}\begin{bmatrix} 0 & 4 \\ 4 & 1 \end{bmatrix}$.

E2

Find the determinant of $\begin{bmatrix} 8 & 6 \\ -7 & 5 \end{bmatrix}$.

E3

Find the inverse of $\begin{bmatrix} 4 & -3 \\ 1 & -2 \end{bmatrix}$.

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Teacher answers & working

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S1

Find $\begin{bmatrix} -2 & 0 \\ 1 & -3 \end{bmatrix}\begin{bmatrix} -3 & 4 \\ 5 & -3 \end{bmatrix}$.

Answer:

$\begin{bmatrix} 6 & -8 \\ -18 & 13 \end{bmatrix}$

Working:

Use row by column multiplication.

Top-left: -2×-3 + 0×5 = 6.

Top-right: -2×4 + 0×-3 = -8.

Bottom-left: 1×-3 + -3×5 = -18.

Bottom-right: 1×4 + -3×-3 = 13.

Result: $\begin{bmatrix} 6 & -8 \\ -18 & 13 \end{bmatrix}$.

S2

Find the determinant of $\begin{bmatrix} -2 & 6 \\ 7 & 5 \end{bmatrix}$.

Answer:

-52

Working:

For a 2 by 2 matrix, det(A) = ad − bc.

$\det(A) = -2\times5 - 6\times7$.

$\det(A) = -52$.

Teacher answers & working

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S3

Find the inverse of $\begin{bmatrix} -4 & 6 \\ 0 & 2 \end{bmatrix}$.

Answer:

$\begin{bmatrix} -1/4 & 3/4 \\ 0 & 1/2 \end{bmatrix}$

Working:

For a 2 by 2 matrix A, use $A^{-1} = 1/(ad - bc) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

First find the determinant: det(A) = -4×2 − 6×0 = -8.

Because the determinant is not 0, the inverse exists.

Form the adjugate by swapping the leading diagonal entries and changing the signs of the other diagonal.

Adjugate matrix: $\begin{bmatrix} 2 & -6 \\ 0 & -4 \end{bmatrix}$.

$A^{-1} = 1/-8 \begin{bmatrix} 2 & -6 \\ 0 & -4 \end{bmatrix}$.

Simplify each fraction if possible.

So $A^{-1} = \begin{bmatrix} -1/4 & 3/4 \\ 0 & 1/2 \end{bmatrix}$.

S4

Find the determinant of $\begin{bmatrix} -1 & 2 & 0 \\ 5 & 5 & -3 \\ 4 & -1 & -4 \end{bmatrix}$.

Answer:

39

Working:

Expand along the first row.

det(A) = -1(5×-4 − -3×-1) − 2(5×-4 − -3×4) + 0(5×-1 − 5×4).

det(A) = 39.

Teacher answers & working

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E1

Find $\begin{bmatrix} -1 & -3 \\ 5 & -1 \end{bmatrix}\begin{bmatrix} 0 & 4 \\ 4 & 1 \end{bmatrix}$.

Answer:

$\begin{bmatrix} -12 & -7 \\ -4 & 19 \end{bmatrix}$

Working:

Use row by column multiplication.

Top-left: -1×0 + -3×4 = -12.

Top-right: -1×4 + -3×1 = -7.

Bottom-left: 5×0 + -1×4 = -4.

Bottom-right: 5×4 + -1×1 = 19.

Result: $\begin{bmatrix} -12 & -7 \\ -4 & 19 \end{bmatrix}$.

E2

Find the determinant of $\begin{bmatrix} 8 & 6 \\ -7 & 5 \end{bmatrix}$.

Answer:

82

Working:

For a 2 by 2 matrix, det(A) = ad − bc.

$\det(A) = 8\times5 - 6\times\text{-}7$.

$\det(A) = 82$.

Teacher answers & working

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E3

Find the inverse of $\begin{bmatrix} 4 & -3 \\ 1 & -2 \end{bmatrix}$.

Answer:
$\begin{bmatrix} 2/5 & -3/5 \\ 1/5 & -4/5 \end{bmatrix}$

Working:

For a 2 by 2 matrix A, use $A^{-1} = 1/(ad - bc) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

First find the determinant: det(A) = 4×-2 − -3×1 = -5.

Because the determinant is not 0, the inverse exists.

Form the adjugate by swapping the leading diagonal entries and changing the signs of the other diagonal.

Adjugate matrix: $\begin{bmatrix} -2 & 3 \\ -1 & 4 \end{bmatrix}$.

$A^{-1} = 1/-5 \begin{bmatrix} -2 & 3 \\ -1 & 4 \end{bmatrix}$.

Simplify each fraction if possible.

So $A^{-1} = \begin{bmatrix} 2/5 & -3/5 \\ 1/5 & -4/5 \end{bmatrix}$.